



IKRGY Infotech Pvt Ltd

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SMART AQUACULTURE MONITORING & MANAGEMENT PLATFORM

IoT-Based Real-Time Water Quality Monitoring and Pond Management System



1. INTRODUCTION

Aquaculture is one of the fastest-growing food production industries in the world. Fish farming and aquatic cultivation play a major role in providing food, employment, and economic growth in many countries. Maintaining proper water quality is one of the most important factors in successful aquaculture management because the health and growth of aquatic organisms completely depend on the surrounding water conditions.

Traditional aquaculture farms mainly rely on manual monitoring methods to check water quality parameters such as pH, temperature, and turbidity. These methods require continuous human effort, consume more time, and often fail to provide accurate real-time monitoring. Sudden changes in water quality can cause stress, diseases, oxygen imbalance, and even large-scale fish mortality, resulting in significant financial losses for farm owners.

With the rapid growth of Internet of Things (IoT) technologies, smart monitoring systems are becoming an effective solution for modern aquaculture farms. IoT-based monitoring systems help in collecting real-time sensor data, transmitting the data remotely, storing historical records, and generating alerts whenever abnormal conditions are detected. These systems reduce manual work and improve operational efficiency in fish farming environments.

The Smart Aquaculture Monitoring & Management Platform is designed to provide an intelligent and automated solution for monitoring pond water quality using ESP32, GSM communication, and multiple sensors. The current prototype focuses on monitoring pH, temperature, and turbidity parameters. The collected sensor data is transmitted to the backend server, processed, and displayed through a centralized dashboard for monitoring and analysis.

In addition to water quality monitoring, the platform also includes features such as device management, pond management, role-based access control, subscription plans, owner management, and agent assignment. The system allows Managers to monitor the complete platform, Owners to manage ponds and subscriptions, and Agents to operate assigned devices in the field.

The proposed system aims to improve aquaculture productivity, reduce fish mortality, simplify farm management, and provide a scalable digital solution for modern aquaculture industries. Future enhancements of the project include ammonia and nitrate monitoring, advanced analytics, AI-based prediction systems, and mobile application integration for smarter aquaculture management.



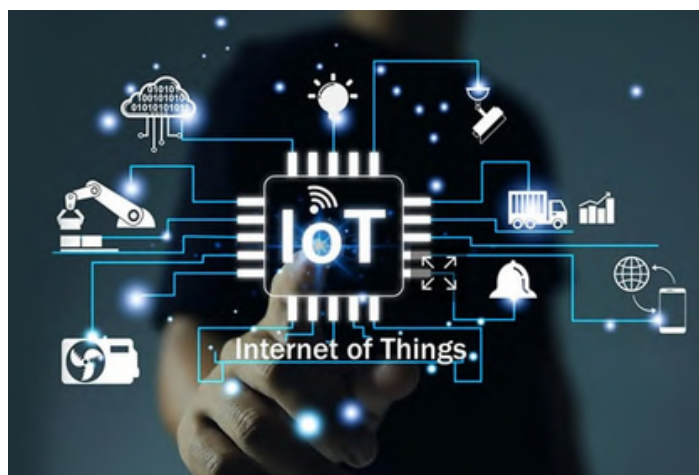
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1.1 Introduction to IoT

The Internet of Things (IoT) is a modern technology that connects physical devices, sensors, and electronic systems to collect and exchange data through communication networks. IoT enables real-time monitoring, automation, and smart decision-making by continuously gathering information from connected devices. In recent years, IoT has become one of the most important technologies in various industries such as healthcare, agriculture, smart cities, industrial automation, and aquaculture. IoT systems help reduce manual work, improve operational efficiency, and provide accurate real-time data for better management and analysis.

In aquaculture, IoT plays a major role in monitoring water quality parameters such as pH, temperature, turbidity, ammonia, and nitrate levels. Maintaining proper water conditions is very important for the healthy growth and survival of aquatic organisms. Traditional monitoring methods are mostly manual and time-consuming, which may lead to delayed responses during critical situations. IoT-based monitoring systems help overcome these problems by continuously collecting sensor data and transmitting it to centralized monitoring platforms for real-time analysis and management.

The proposed Smart Aquaculture Monitoring & Management Platform uses IoT technology with ESP32, GSM communication, and multiple sensors to monitor pond conditions in real time. The system helps farm owners and agents remotely monitor water quality, manage ponds, receive alerts, and improve overall aquaculture productivity through smart digital monitoring solutions.





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1.2 Introduction to Full Stack Development

Full stack development is the process of developing both the frontend and backend components of a software application. A full stack system includes user interface design, server-side logic, database management, API development, authentication, and communication between hardware and software systems. Full stack development helps in creating complete end-to-end applications that are scalable, secure, and capable of handling real-time data processing and management.

The frontend of an application is responsible for displaying data and providing interactive dashboards for users. It helps users monitor information, manage devices, view analytics, and perform system operations through a user-friendly interface. The backend handles the core functionalities of the application such as data processing, business logic, API management, authentication, and database operations. The database stores all important information including user details, sensor readings, device information, pond management data, subscriptions, and historical reports.

In the proposed Smart Aquaculture Monitoring & Management Platform, full stack development is used to build a complete monitoring and management system for aquaculture farms. The frontend dashboard is developed using React.js for real-time visualization and management, while the backend is developed using FastAPI for handling APIs, device communication, authentication, and data processing. PostgreSQL is used as the database for secure data storage and management. The full stack architecture enables Managers, Owners, and Agents to access the platform, monitor water quality parameters, manage ponds and devices, and analyze real-time sensor data efficiently through a centralized digital system.





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2. Problem Statement

Aquaculture farming requires continuous monitoring of water quality parameters to maintain healthy environmental conditions for fish and aquatic organisms. Important parameters such as pH, temperature, and turbidity directly affect fish growth, feeding behavior, oxygen balance, and overall pond health. In traditional aquaculture systems, these parameters are mostly monitored manually, which is time-consuming, labor-intensive, and often inaccurate. Manual monitoring methods do not provide real-time data or immediate alerts during sudden changes in water conditions, which can lead to fish stress, diseases, poor water quality, and high mortality rates.

Another major challenge in aquaculture management is the lack of centralized monitoring and digital management systems. Farm owners managing multiple ponds often face difficulties in tracking water conditions, managing devices, assigning workers, and analyzing pond performance efficiently. Existing systems generally lack real-time analytics, role-based management, automated alerts, and remote accessibility. Additionally, there is no proper mechanism to manage subscription-based monitoring services or provide parameter-based access control for different users.

Field agents responsible for monitoring ponds also face operational difficulties due to the absence of device-level access management and pond assignment systems. Without proper tracking and monitoring infrastructure, it becomes difficult to ensure accurate pond-wise data collection and management activities.

Therefore, there is a need for a smart, scalable, and cost-effective aquaculture monitoring platform that can provide real-time water quality monitoring, GSM-based remote communication, centralized dashboard management, role-based access control, pond management, subscription handling, and intelligent analytics to improve productivity and reduce losses in modern aquaculture farming.



3. Existing System

The existing aquaculture monitoring system mainly relies on manual methods and basic monitoring tools to observe water quality parameters such as pH, temperature, and turbidity. Farmers or field workers are required to physically visit ponds and measure these parameters using separate instruments. This approach is time-consuming, labor-intensive, and does not provide continuous monitoring of water conditions.

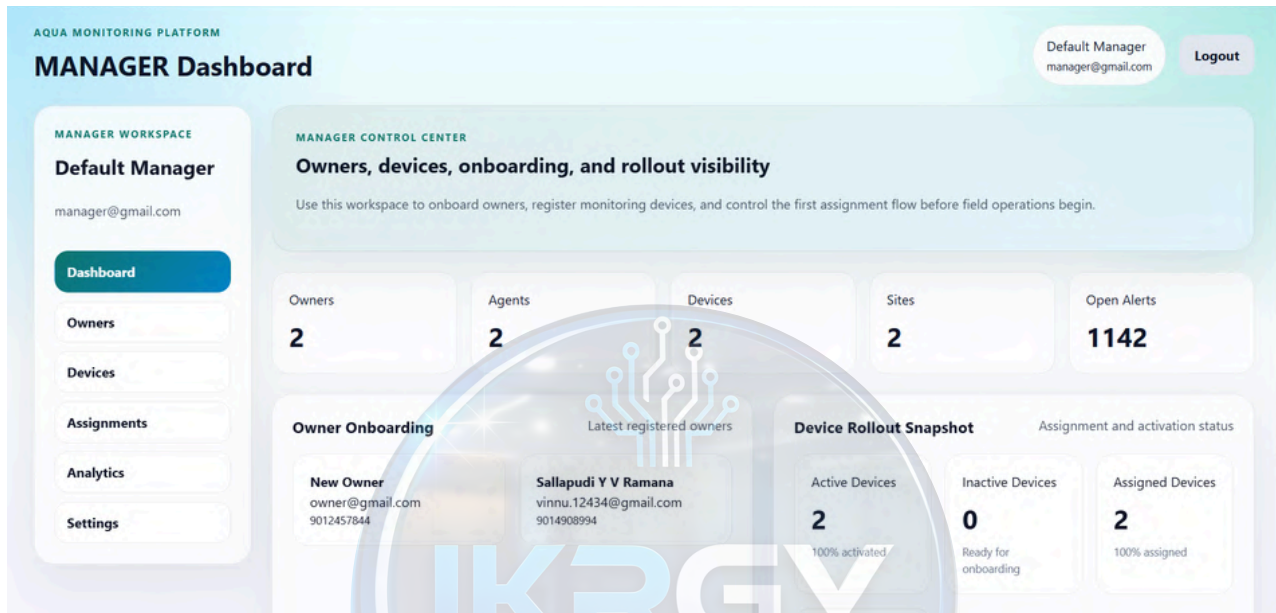
In the current system, there is no real-time data transmission or centralized monitoring platform. The collected data is not stored digitally in most cases, making it difficult to analyze historical trends or make data-driven decisions. Due to the lack of automated alerts, sudden changes in water quality often go unnoticed, which can negatively impact fish health and increase the risk of losses.

Additionally, there is no structured management system to handle multiple ponds, devices, or field workers efficiently. Farm owners find it difficult to track operations, assign responsibilities, and monitor activities remotely. The absence of role-based access, device management, and subscription-based services limits the scalability and efficiency of traditional aquaculture monitoring systems.

Overall, the existing system lacks automation, real-time monitoring, centralized control, and efficient management features, highlighting the need for a more advanced and integrated aquaculture monitoring solution.



3.1 Manager Dashboard



The Manager Dashboard is the central administrative module of the Smart Aquaculture Monitoring & Management Platform. It provides complete visibility and control over the entire system, including owners, devices, subscriptions, assignments, onboarding activities, and operational analytics. The dashboard helps Managers efficiently monitor and manage all aquaculture operations through a centralized digital interface.

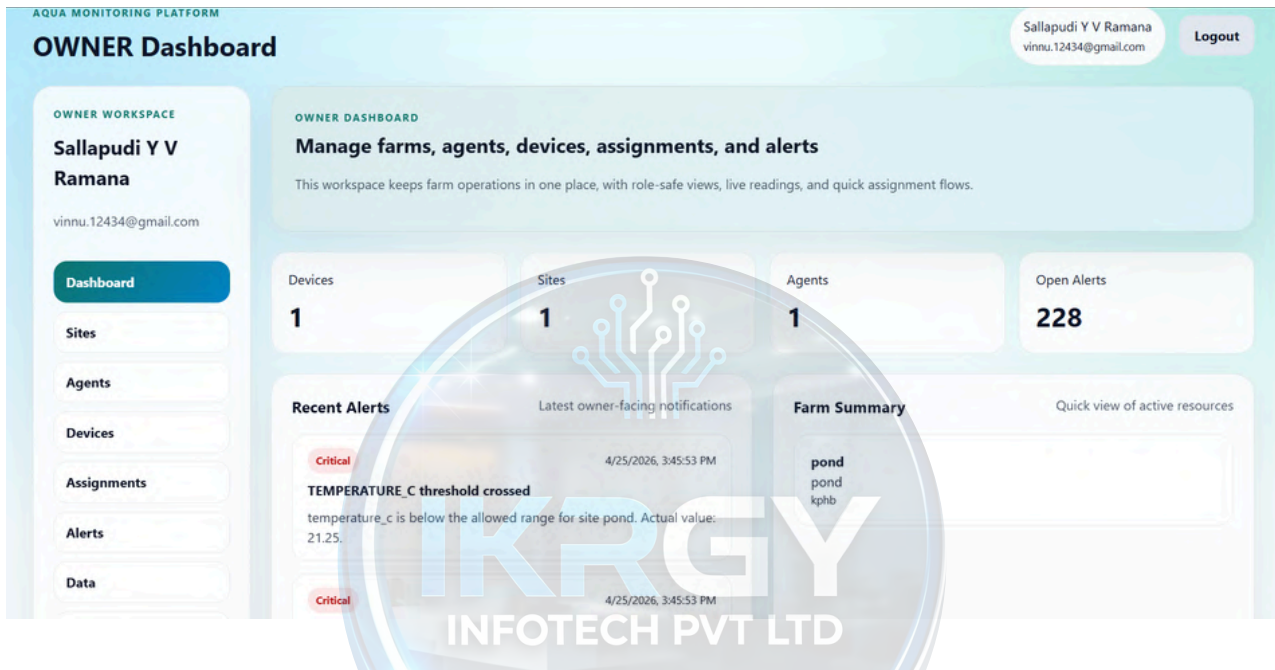
The Manager is responsible for creating and managing Owners within the platform. The dashboard allows Managers to onboard new Owners, monitor registered users, manage devices, and track subscription activities. Managers can also monitor active and inactive devices, assignment status, and overall rollout activities before field operations begin.

The Manager Dashboard also provides real-time analytics such as the total number of Owners, Agents, Devices, Sites, and Open Alerts available in the system. These analytics help Managers quickly understand operational status, monitor system performance, and manage aquaculture monitoring activities efficiently through a secure and user-friendly dashboard.



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3.2 Owner Dashboard



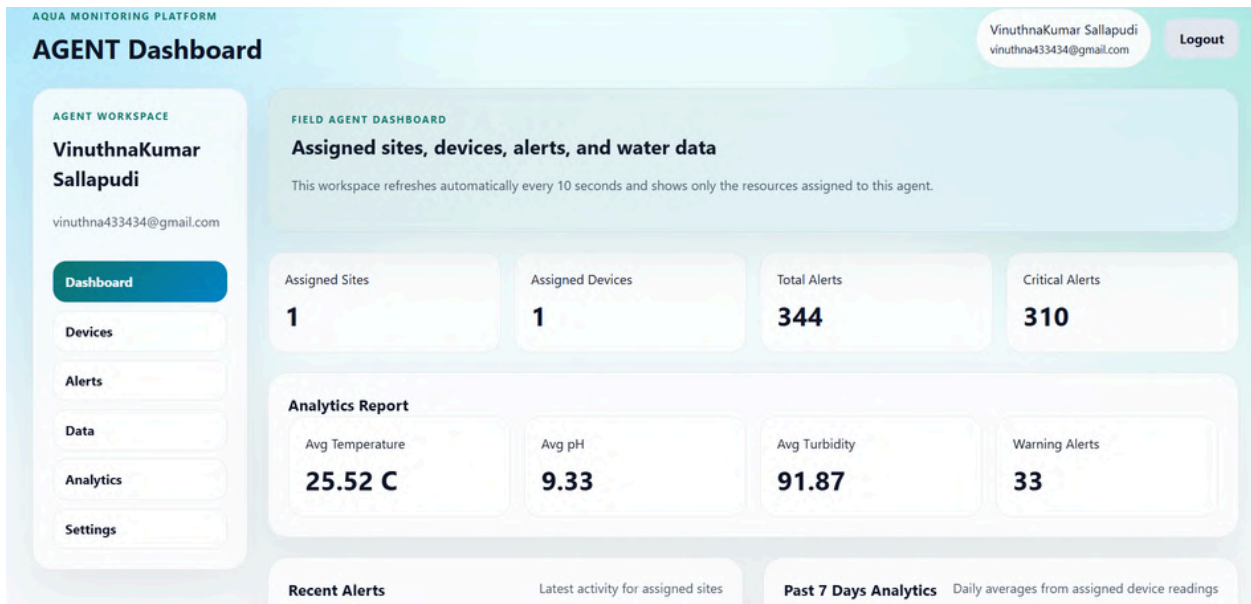
The Owner Dashboard is designed to help aquaculture farm owners manage their complete monitoring operations through a centralized digital platform. The dashboard provides real-time visibility of ponds, devices, agents, assignments, alerts, and water quality monitoring activities. Owners can efficiently monitor farm operations and access all important management functionalities from a single interface.

The dashboard allows Owners to manage ponds, create and assign Agents, monitor connected devices, and track live alerts generated from water quality monitoring systems. Owners can also view operational summaries such as the number of active devices, sites, agents, and critical alerts within the system. This helps improve farm management efficiency and enables quick response during abnormal pond conditions.

The Owner Dashboard also supports role-based access and subscription-based monitoring services. Owners can access monitoring features based on their selected subscription plans and parameters. The dashboard provides secure access to monitoring data, assignment management, alert tracking, and analytical information required for effective aquaculture farm management.



3.3 Agent Dashboard



The Agent Dashboard is designed to support field-level operations by providing access to assigned ponds, devices, alerts, and real-time water quality data. It allows agents to view only the resources that are specifically assigned to them, ensuring controlled and secure access within the system. This helps maintain proper monitoring and operational discipline at the field level.

The dashboard provides information such as assigned sites, assigned devices, total alerts, and critical alerts related to water quality conditions. Agents can also view basic analytics including average temperature, pH levels, turbidity values, and warning alerts, which help them understand the current status of the pond and take necessary actions when required.

Before performing any monitoring activity, the agent is required to activate the device and select the assigned pond. This ensures that all data is properly mapped to the correct pond and device. The Agent Dashboard simplifies field operations by providing a clear view of assigned tasks, real-time alerts, and essential monitoring data, enabling efficient and accurate aquaculture management at the ground level.



4. Technologies and Components Used (Existing System)

The existing system is developed using a combination of IoT hardware components, frontend technologies, backend frameworks, and database systems to enable complete aquaculture monitoring and management.

IoT Components

- ESP32 Microcontroller
- pH Sensor (Analog pH Sensor Module)
- Temperature Sensor (DS18B20)
- Turbidity Sensor
- GSM Module (SIM800L / SIM900A)
- Power Supply (12V Battery)
- Buzzer (for alerts)
- Connecting wires, resistors, and breadboard setup

Backend Technologies

- FastAPI (Python framework for API development)
- REST APIs for communication between device and server
- Authentication & Authorization (JWT-based login system)
- Data processing and alert generation logic

Frontend Technologies

- React.js (for building dashboards)
- HTML, CSS, JavaScript
- Responsive UI for Manager, Owner, and Agent dashboards
- Data visualization components for analytics and monitoring

Database

PostgreSQL (for storing user data, device data, sensor readings, alerts, and subscriptions)



5. Feature Enhancements

The system can be further improved by integrating additional advanced features to enhance monitoring accuracy, operational efficiency, and overall system intelligence. These enhancements focus on improving water quality analysis, strengthening field-level operations, and providing better decision-making capabilities through advanced data insights.

One of the key enhancements is the integration of an Ammonia Sensor, which helps in detecting ammonia levels in water. Ammonia is a critical parameter in aquaculture, and high levels can be harmful to aquatic life. Similarly, the addition of a Nitrate Sensor allows better monitoring of water chemistry, ensuring a balanced aquatic environment and improving fish health management.

Another important enhancement is the development of an advanced alert system that provides intelligent notifications based on threshold values. This system can help users receive timely alerts for abnormal conditions, enabling quick action to prevent losses. Additionally, a dedicated Agent Device Interface can be introduced, where agents can securely log in directly through the device, select assigned ponds, and perform monitoring activities more efficiently.

The system can also be enhanced by improving analytics and reporting features. Advanced analytics can provide detailed insights into historical data, trends, and performance metrics of water quality parameters. These insights will help farm owners and managers make better decisions and optimize aquaculture operations.

Overall, these feature enhancements aim to transform the existing system into a more intelligent, scalable, and efficient smart aquaculture monitoring platform by improving monitoring capabilities, user experience, and data-driven decision-making.



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5.1 Ammonia Sensor

The integration of an Ammonia Sensor enhances the aquaculture monitoring system by enabling continuous measurement of ammonia levels in pond water. Ammonia is a highly critical parameter, as elevated levels can be toxic to fish and other aquatic organisms, leading to stress, reduced growth, and even mortality. By incorporating ammonia monitoring, the system provides a more comprehensive understanding of water quality conditions.

With this enhancement, the system can detect abnormal increases in ammonia levels and generate alerts to notify users immediately. This allows farm owners and agents to take timely corrective actions such as water exchange or treatment, preventing potential damage to aquatic life. The addition of ammonia monitoring significantly improves the reliability and effectiveness of the overall aquaculture management system.





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An ammonia sensor is a device used to measure the concentration of ammonia present in water. It works by detecting chemical changes in the water and converting them into electrical signals that can be processed by a microcontroller. Ammonia is a critical parameter in aquaculture, as even small increases in its concentration can be harmful to fish and other aquatic organisms.

In aquaculture environments, ammonia is mainly produced from fish waste, uneaten feed, and organic decomposition in the pond. If ammonia levels are not properly monitored and controlled, it can lead to stress, reduced growth, diseases, and even death of aquatic species. Therefore, continuous monitoring of ammonia is very important to maintain a healthy and balanced aquatic environment.

In this project, the ammonia sensor is integrated with the monitoring system to provide real-time measurement of ammonia levels in pond water. The collected data is transmitted to the system, where it can be analyzed and monitored through the dashboard. When ammonia levels exceed safe limits, the system can generate alerts to notify users, enabling quick corrective actions such as water exchange or treatment.

By incorporating the ammonia sensor, the system becomes more reliable and efficient in water quality monitoring. This enhancement helps improve fish health, reduce risks, and support better decision-making for aquaculture management.



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5.2 Nitrate Monitoring Enhancement

The integration of a Nitrate Sensor enhances the aquaculture monitoring system by enabling continuous measurement of nitrate levels in pond water. Nitrate is an important water quality parameter that indicates the level of nutrient concentration and overall water health. High nitrate levels can lead to poor water conditions, affecting fish growth and increasing the risk of diseases.

With this enhancement, the system can monitor nitrate levels in real time and provide alerts when the values exceed safe limits. This helps farm owners and agents take necessary actions such as water treatment or pond maintenance to maintain a balanced aquatic environment.

By incorporating nitrate monitoring, the system provides a more complete analysis of water quality along with other parameters like pH, temperature, turbidity, and ammonia. This improves the accuracy, efficiency, and reliability of the aquaculture monitoring system.





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The integration of a nitrate sensor improves the aquaculture monitoring system by enabling continuous measurement of nitrate levels in pond water. Nitrate is an important water quality parameter that reflects nutrient levels and overall pond health. While it is less toxic than ammonia, excessive nitrate accumulation over time can negatively affect fish growth, reduce water quality, and lead to long-term health issues in aquatic organisms.

In aquaculture environments, nitrate is mainly formed as a result of the breakdown of ammonia through biological processes. This makes it an important indicator of the nitrogen cycle within the pond. Monitoring nitrate levels helps in understanding how effectively the pond ecosystem is maintaining balance and whether any corrective actions are required to maintain optimal conditions.

In this project, the nitrate sensor is integrated into the system to provide real-time monitoring of nitrate concentration. The sensor data is collected and processed by the system and displayed on the dashboard for easy analysis. When nitrate levels exceed safe limits, alerts can be generated to notify users, enabling timely actions such as water exchange or improved pond management practices.

By adding nitrate monitoring along with other parameters like pH, temperature, turbidity, and ammonia, the system becomes more comprehensive and reliable. This enhancement supports better decision-making, improves water quality management, and helps ensure a healthier environment for aquaculture operations.

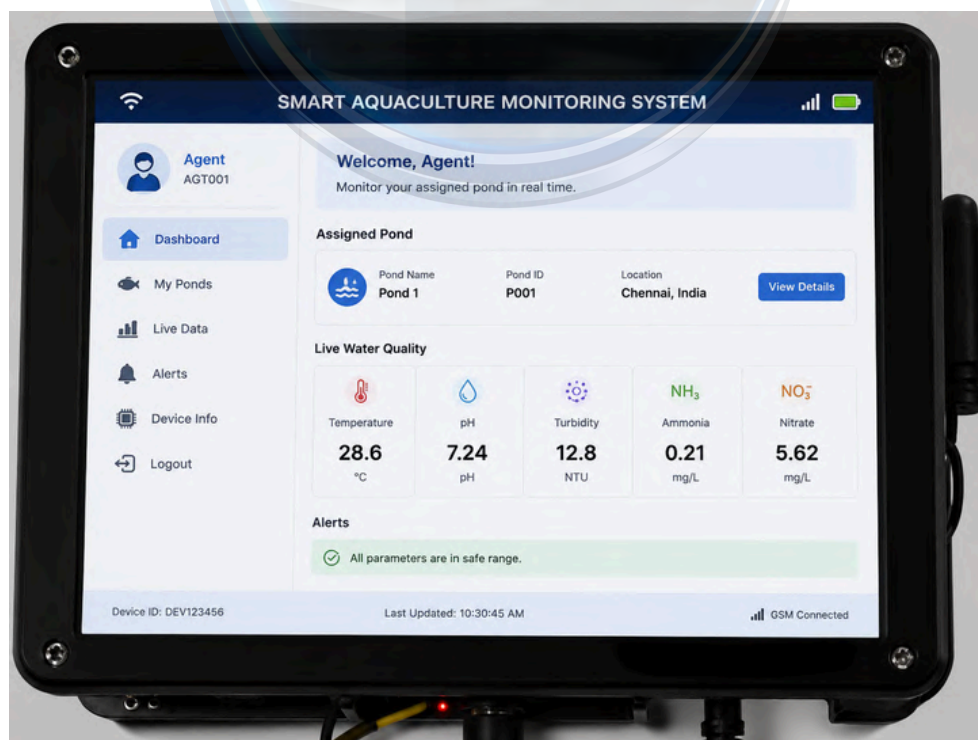


5.3 Agent Device Interface Enhancement

The Agent Device Interface enhancement focuses on improving field-level operations by providing a dedicated interface for agents to interact directly with the monitoring device. This feature allows agents to securely access the system through the device, ensuring that only authorized users can operate and monitor assigned ponds.

With this enhancement, agents can log in using their credentials, select the assigned pond, and begin monitoring activities. This ensures that all data collected from the sensors is accurately mapped to the correct pond and device, reducing errors and improving data reliability. The interface also helps agents view basic information such as device status and alerts while working in the field.

This enhancement simplifies the overall workflow for agents by providing a structured and controlled process for device usage. It improves operational efficiency, ensures proper data tracking, and enhances the accuracy of aquaculture monitoring at the ground level.

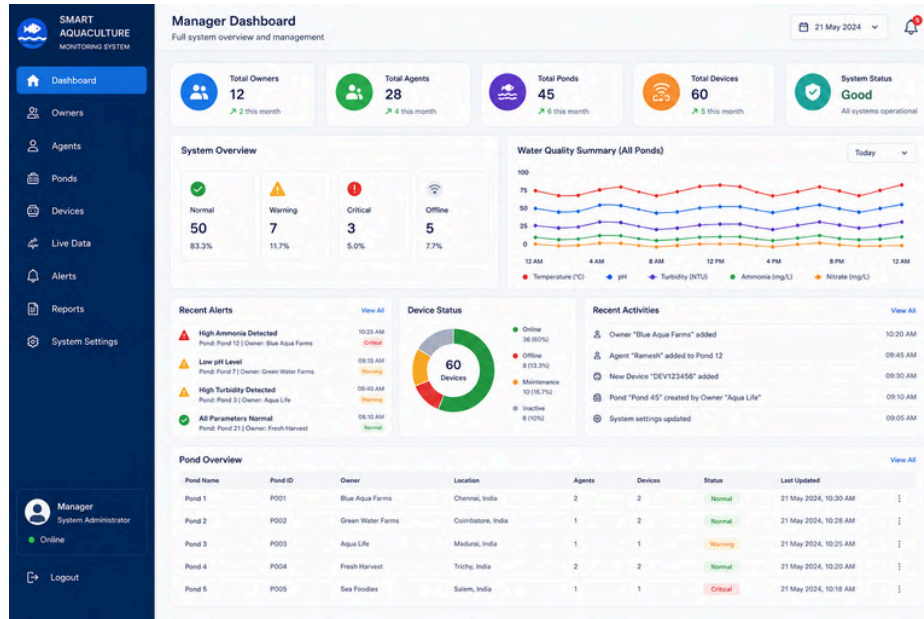




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Manager Dashboard:



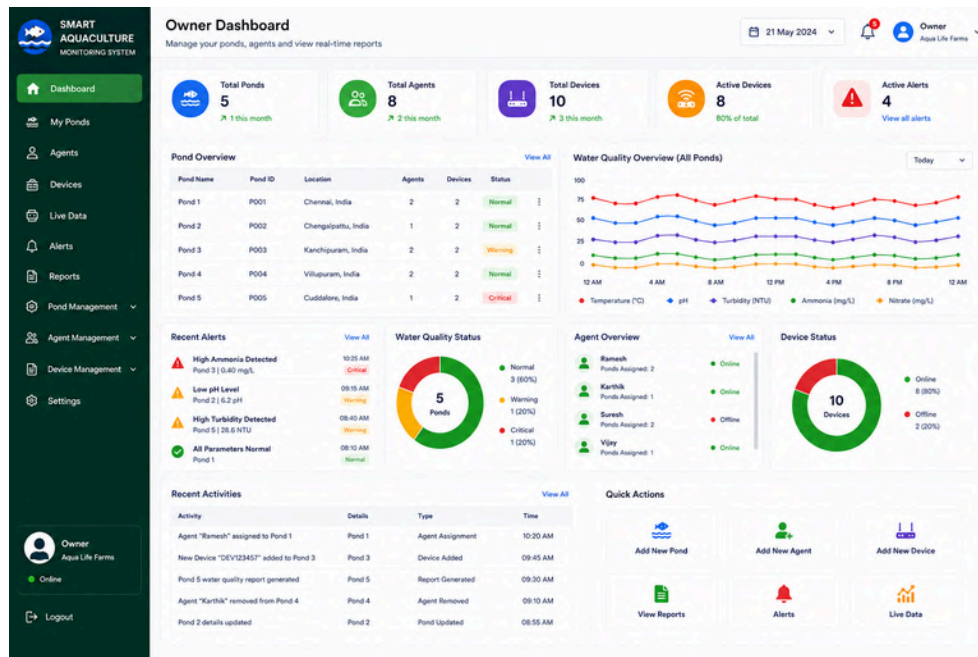
The Manager Dashboard is designed to provide a centralized view of all aquaculture operations, enabling efficient monitoring and decision-making. It displays real-time data from multiple ponds, including key water quality parameters such as temperature, pH, turbidity, ammonia, and nitrate levels. The dashboard highlights the overall system status using visual indicators, helping managers quickly identify whether conditions are normal or require attention. With summarized insights and alerts, the manager can easily track the health and performance of all connected devices and ponds.

In addition to real-time monitoring, the dashboard offers analytical features such as trend graphs and historical data comparison. These insights help managers understand patterns, detect anomalies, and make data-driven decisions to improve productivity and reduce risks. The system also provides alerts and notifications for abnormal conditions, ensuring timely intervention to prevent potential losses. Device status, connectivity, and last updated timestamps are clearly displayed to maintain operational transparency.

Furthermore, the Manager Dashboard supports efficient management of agents and field operations. It allows tracking of assigned ponds, monitoring agent activities, and ensuring that all devices are functioning properly. By integrating backend systems and IoT devices, the dashboard acts as a bridge between field data and strategic planning, ultimately enhancing operational efficiency and improving overall aquaculture management.



Owner Dashboard



The Owner Dashboard provides a high-level overview of the entire aquaculture operation, focusing on business performance, system health, and overall productivity. It presents summarized data from all ponds and locations, including key water quality parameters and operational status. The dashboard uses clear visual indicators and reports to help the owner quickly understand whether the system is running efficiently without needing to monitor every detail.

In addition to monitoring, the Owner Dashboard emphasizes insights and reporting. It includes analytics such as overall production trends, water quality summaries, and performance comparisons across different ponds or regions. This helps the owner make strategic decisions, optimize resources, and plan for future improvements. Alerts and notifications are also available to highlight critical issues that may impact business outcomes.

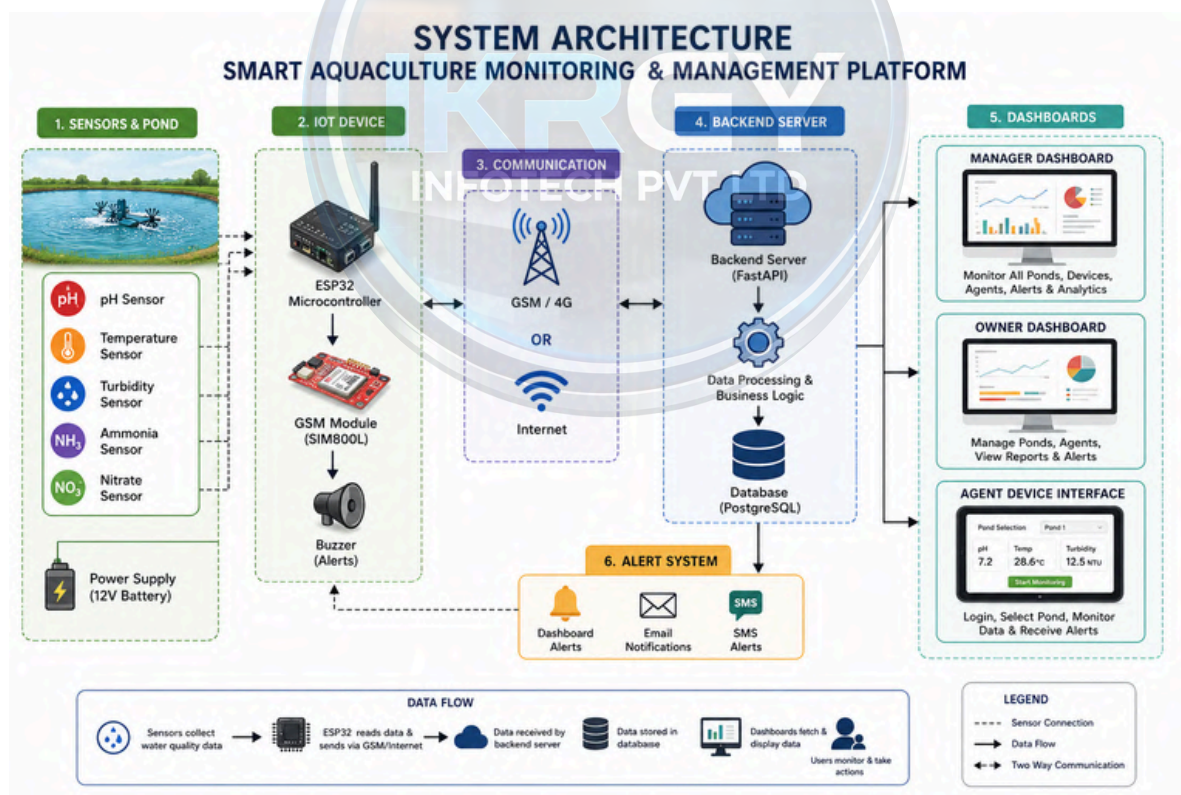
Furthermore, the dashboard enables oversight of managers, agents, and system usage. It provides visibility into how operations are being handled at different levels, ensuring accountability and smooth coordination. By integrating IoT data with backend analytics, the Owner Dashboard supports better decision-making, improved efficiency, and long-term growth of the aquaculture system.



6. System Architecture :

The Smart Aquaculture Monitoring & Management Platform follows a multi-layer architecture that integrates IoT devices, backend systems, and user dashboards. The system consists of sensors connected to a microcontroller (ESP32), which collects real-time water quality data such as pH, temperature, turbidity, ammonia, and nitrate levels. The collected data is transmitted to the backend server using GSM or internet-based communication.

The backend system processes the incoming data, stores it in the database, and generates alerts when abnormal conditions are detected. The processed data is then displayed on role-based dashboards including Manager, Owner, and Agent interfaces. This architecture ensures real-time monitoring, centralized management, and efficient decision-making.

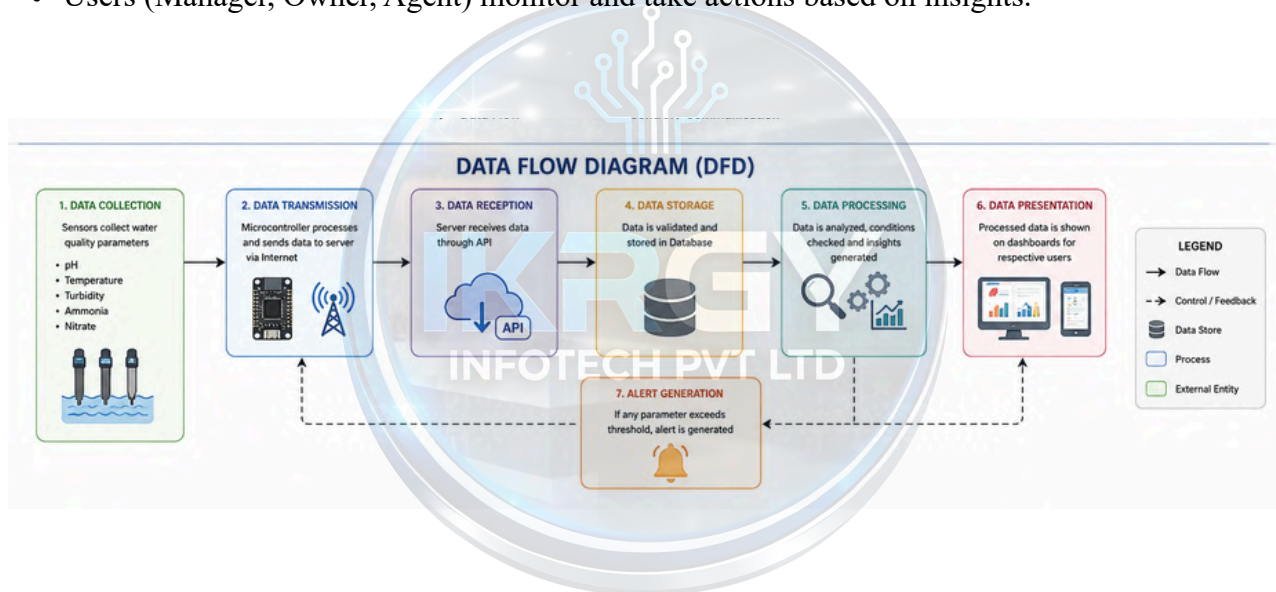




7. Data Flow Diagram (DFD)

The data flow in the system occurs in the following steps:

- Sensors collect water quality data from ponds.
- The microcontroller processes and formats the data.
- Data is transmitted to the backend server via APIs.
- The backend stores the data in the database.
- The system analyzes the data and generates alerts if required.
- Dashboards fetch data from the backend and display it to users.
- Users (Manager, Owner, Agent) monitor and take actions based on insights.





8. Database Design

The system uses a structured database to store and manage all information.

Tables :

1. Users Table

- user_id
- name
- email
- password
- role (Manager / Owner / Agent)

2. Ponds Table

- pond_id
- pond_name
- location
- owner_id

3. Devices Table

- device_id
- pond_id
- status
- last_active

4. Sensor Data Table

- data_id
- pond_id
- temperature
- pH
- turbidity
- ammonia
- nitrate
- timestamp





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5. Alerts Table

- alert_id
- pond_id
- parameter
- value
- status
- timestamp

6. Assignments Table

- assignment_id
- agent_id
- pond_id





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9. API Design

The system uses REST APIs for communication between devices and backend.

Sample APIs:

1. Login API

- Endpoint: /api/login
- Method: POST
- Request:

```
{  
  "email": "user@gmail.com",  
  "password": "123456"  
}
```

2. Send Sensor Data

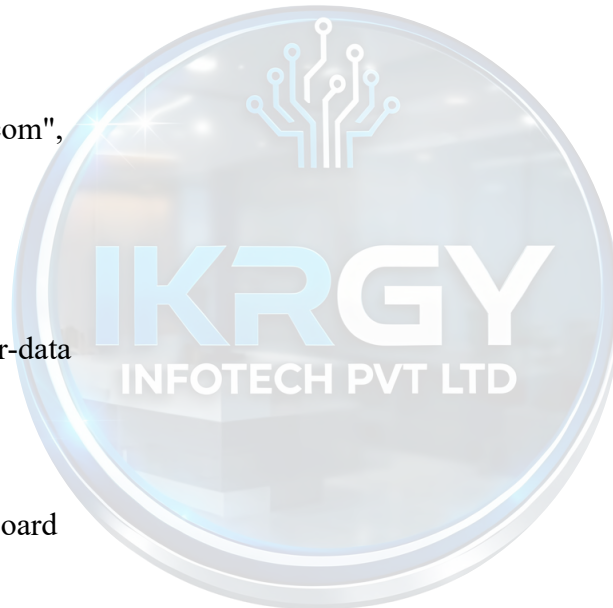
- Endpoint: /api/sensor-data
- Method: POST

3. Get Dashboard Data

- Endpoint: /api/dashboard
- Method: GET

4. Alerts API

- Endpoint: /api/alerts
- Method: GET





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10. User Roles and Permissions

Role	Permissions
Manager	Manage owners, devices, and system analytics
Owner	Manage ponds, agents, and view reports
Agent	Access assigned ponds and operate devices





11. Advantages of the Proposed System

Real-Time Monitoring and Visibility

The system provides live tracking of agent activities and device data, enabling managers and owners to monitor operations instantly and make quick decisions.

Centralized Control System

All data and functionalities are integrated into a single platform, making it easy to manage agents, devices, and reports without using multiple systems.

Role-Based Access Control

The system offers separate dashboards for Owner, Manager, and Agent, ensuring secure access and a customized user experience based on roles.

Improved Productivity

Automation and simplified workflows help agents complete tasks efficiently, while managers can easily track performance, reducing manual effort.

Data-Driven Decision Making

The system provides detailed analytics and reports, helping management make informed decisions based on real-time data and performance metrics.

Enhanced Communication

It enables smooth communication between agents, managers, and owners through task updates, alerts, and notifications.

Automation and Alerts

Automatic notifications and alerts reduce the need for manual monitoring and ensure timely responses to important events.

Secure Data Management

The system ensures data security through authentication and access control, preventing unauthorized usage and protecting sensitive information.



12. Conclusion

The proposed system successfully integrates the functionalities of Owner, Manager, and Agent into a unified and efficient platform. By introducing role-based dashboards and a dedicated device interface for agents, the system ensures smooth coordination, real-time monitoring, and effective management of operations.

The implementation of centralized control and real-time data tracking significantly enhances transparency and allows higher authorities to make informed decisions quickly. Managers can efficiently supervise tasks and monitor agent performance, while owners gain a comprehensive overview of the entire system through detailed analytics and reports.

Moreover, the system improves productivity by reducing manual work, minimizing errors, and enabling faster communication between all users. The inclusion of automation features such as alerts and notifications further strengthens the responsiveness and reliability of the system.

The scalability and flexibility of the proposed solution make it suitable for future expansion, allowing the integration of additional features, users, and devices as needed. Security measures such as role-based access control ensure that sensitive data remains protected.

In conclusion, the system provides a modern, efficient, and user-friendly solution for managing operations, improving overall performance, and supporting data-driven decision-making. It serves as a strong foundation for future enhancements and technological advancements.